

🥗 AI-Powered Nutrition Agent

****Personalized Dietary Guidance System built on IBM watsonx.ai & Granite Models****

"Exploring the Power of Agentic AI with IBM Cloud and Orchestrate"

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📌 Overview

The ****AI-Powered Nutrition Agent**** is a multi-agent, agentic AI system that delivers personalized, real-time dietary guidance for individuals, families, and health professionals. It combines ****Retrieval-Augmented Generation (RAG)**** with ****IBM watsonx Orchestrate**** and ****IBM Granite models**** to move beyond static diet advice into an adaptive, context-aware digital nutritionist.

The Problem

Good nutrition is the foundation of health, but people struggle with diet planning, portion control, and tracking intake — and reliable, personalized guidance is scattered and hard to find.

The Solution

An intelligent nutrition assistant that recommends meals, retrieves verified nutrition data, tracks food logs, and gives preventive health advice — all through natural conversation.

✨ Features

- ****Personalized Diet Plans**** — Meal recommendations based on age, health conditions, allergies, cultural preferences, and fitness goals
- ****RAG-Based Food Data Retrieval**** — Fetches and summarizes accurate nutritional values (calories, macros, vitamins) from trusted USDA & WHO sources
- ****Diet Tracking & Feedback**** — Log meals via text, voice, or images and get instant nutritional analysis

- **Preventive Health Focus** — Disease-specific diet suggestions (e.g., diabetes-friendly, heart-healthy plans)
- **Multi-Modal Input** — Supports text, voice, and image-based meal logging
- **Real-Time Adaptation** — Continuously updates suggestions based on daily meal logs
- **Nutritional Dashboard** — Visualizes daily/weekly calorie and macro intake with AI-generated suggestions

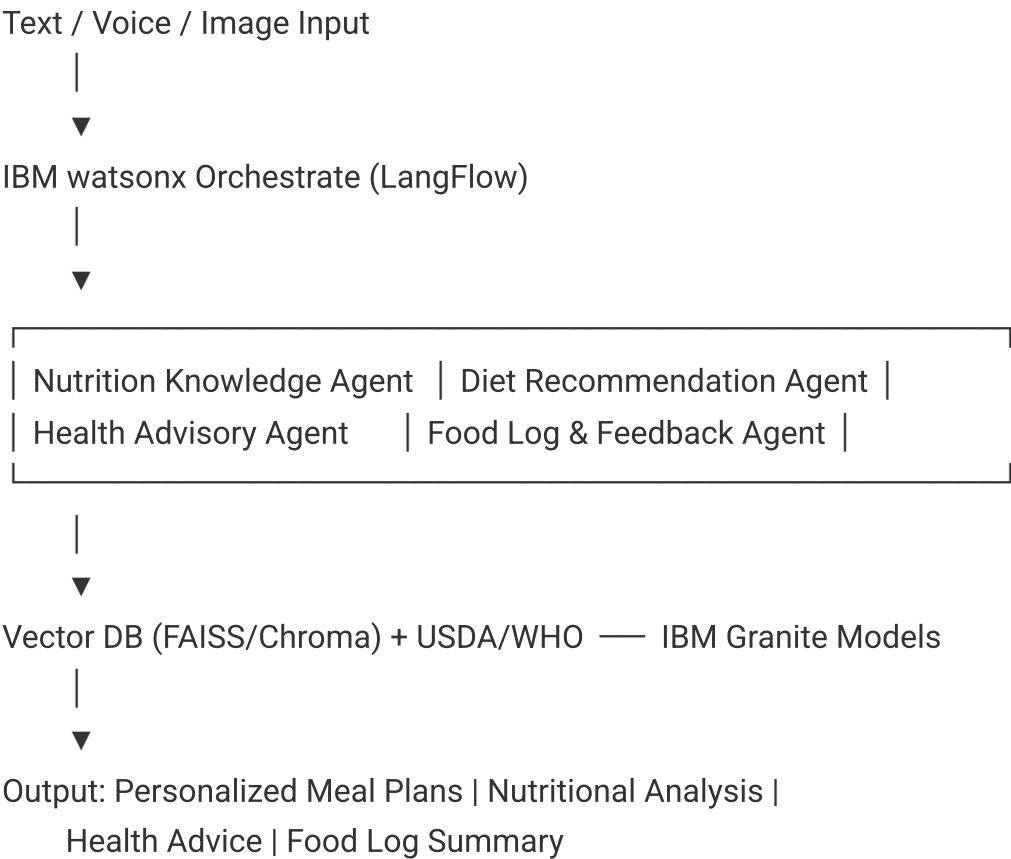
🧠 Multi-Agent Architecture

The system uses four specialized agents orchestrated together rather than a single chatbot:

Agent	Responsibility
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Nutrition Knowledge Agent	Retrieves and summarizes nutritional information using RAG
Diet Recommendation Agent	Generates personalized meal plans from user inputs (age, conditions, preferences, goals)
Health Advisory Agent	Provides preventive health suggestions and disease-specific diets
Food Log & Feedback Agent	Logs meals via text/voice/image and delivers instant nutritional feedback

Architecture Blueprint

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****Agentic AI capabilities:****

- ****Context Awareness**** — understands user age, health conditions, and fitness goals
- ****Real-Time Adaptation**** — updates suggestions based on daily meal logs
- ****Inter-Agent Communication**** — agents collaborate for better decisions
- ****Goal-Driven Behavior**** — proactive and personalized, not just reactive

🛠️ Tech Stack

Component	Purpose
IBM Granite Models (`granite-3.2-8b-instruct`)	Natural language understanding, reasoning, personalization
IBM watsonx.ai	Model deployment, embeddings, AI governance
IBM watsonx Orchestrate	Agent orchestration and workflow management
IBM Cloud (Lite)	Scalable, secure infrastructure for deployment
LangFlow	Orchestration pipeline for agent workflows
RAG (Retrieval-Augmented Generation)	Fetches verified real-time nutrition data before generation
Vector Database (FAISS / Chroma)	Stores/indexes USDA Food Data Central & WHO guidelines for semantic search
Streamlit / React	Dashboard visualizing daily & weekly nutrient intake

watsonx Orchestrate Components Used

1. Chat Input Component — text/voice/image query entry
2. watsonx Orchestrate AI Agent — processes input via Granite models
3. Granite Model (`granite-3.2-8b-instruct`) — fast, real-time diet reasoning
4. Chat Output Component — displays diet plans, analysis, and suggestions
5. Knowledge Base / File Component — stores user data, meal logs, nutrition datasets
6. LangFlow Orchestration Pipeline — ties all agents together

🚀 Quick Start

Prerequisites

- IBM Cloud account (Lite tier is sufficient)
- Access to IBM watsonx.ai / watsonx Orchestrate
- Python 3.9+ (if running the dashboard/RAG pipeline locally)

Setup

1. **Clone the repository**

```
```bash
git clone https://github.com/yashwanthg854-coder/NutritionAgent-IBM-SkillsBuild.git
cd NutritionAgent-IBM-SkillsBuild
```
```

2. **Set up IBM watsonx Orchestrate**

- Create a new Agent in watsonx Orchestrate
- Set the **Agent Behavior Guidelines** (see `/docs` or presentation for the exact prompt)
- Add a **Knowledge Source**: upload USDA/WHO nutrition datasets (PDF, DOCX, TXT, or CSV) or connect a Vector Database (FAISS/Chroma)
- Select the Granite model: `granite-3.2-8b-instruct`
- Save the agent

3. **(Optional) Run the dashboard locally**

```
```bash
pip install -r requirements.txt
streamlit run app.py
```
```

4. **Try it out**

Ask the agent things like:

- **"Suggest a diabetic-friendly dinner"**
- **"Calculate calories for my meal"**
- **"Create a 7-day weight-loss plan"**
- **"Create a vegetarian weight-gain meal plan with calorie targets and macro splits"**

Example Output

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Nutrition Agent:

Here's your 7-Day Vegetarian Weight Gain Plan:

Daily Calorie Target: 2,800–3,000 kcal

Protein: 140g | Carbs: 350g | Fats: 90g

Day 1 – Sample Menu:

Breakfast: Oats + banana + peanut butter smoothie (620 kcal)

Lunch: Paneer tikka + brown rice + dal (780 kcal)

Snack: Mixed nuts + Greek yogurt (310 kcal)

Dinner: Tofu stir-fry + quinoa + avocado salad (730 kcal)

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🌟 Novelty & Uniqueness

- **Multi-Agent Intelligence** instead of a single chatbot
- **RAG-Based Accuracy** reducing misinformation
- **Hyper-Personalization** based on health, culture, and goals
- **Real-Time Feedback Loop** from daily food logs
- **Multi-Modal Input** (text, voice, image)
- **Preventive Healthcare Focus**, not just diet planning
- **Enterprise-ready**, built on IBM watsonx.ai & Granite

🧠 Future Scope

1. **Wearables & Health Device Integration** — real-time steps, heart rate, sleep data
2. **AI Voice Assistant & Multilingual Support** — broader accessibility
3. **Predictive Health Analytics** — long-term dietary pattern analysis for disease prevention
4. **Community & Social Integration** — shared meal plans, recipes, and peer motivation

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🏠 Program

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